A Validated Learned Market Model for Sample-Efficient Battery Storage Arbitrage

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***Abstract*—Training a reinforcement-learning controller for battery-storage arbitrage model-free demands years of once-a-day market data, and any policy tuned on historical prices must eventually face an unseen regime. We exploit a structural property of the problem. A battery small relative to the grid is a *price-taker*, so its environment decomposes into battery dynamics known exactly and an exogenous market price process observed for free, once per day, regardless of policy behaviour. We therefore learn only the market, as an ensemble of hour-autoregressive networks admitted into training only after passing a statistical validation gate against held-out simulated data, and train a Soft Actor-Critic agent Dyna-style inside it, its replay buffer seeded by planner demonstrations so the physical asset never executes an untrained action. Under a fixed real-data budget, this gated model nearly matches an unlimited-data model-free ceiling and clearly beats a replay-only control given identical data and gradient budget. What buys the sample efficiency is the quality of the model, not the quantity of anything: tripling the budget to three years fails the same gate that one year passes, and a replay control given the identical year with unlimited gradient epochs finishes 32 points behind. The gate scores the model against a held-out year from the same simulator, so the standard it enforces is internal calibration rather than resemblance to a measured market. We then map three ways this decomposition stops paying. The policy degrades gracefully, a floor rather than a solution, under unseen transient shocks. A nightly refit-and-replan adaptation loop fails to beat a frozen pre-shift policy under persistent regime shifts, because incremental refitting drives its market model outside the validation gate within three nights, on stationary data and weeks before the shift arrives. And a physics-informed battery surrogate is simply not worth it on this problem, losing to a plain network at every realistic data budget, with an ablation against deliberately wrong physics showing its residual carries no information at those budgets. All results are normalized against exact optimization oracles in a calibrated, deterministic merit-order market simulator enabling exact paired counterfactual evaluation.**

***Index Terms*—reinforcement learning, Dyna, model-based reinforcement learning, battery energy storage, electricity price arbitrage, non-stationarity, physics-informed neural networks**

I. Introduction

A. Background and Motivation

The global transition away from fossil fuels and towards sustainable energy sources is reshaping how modern power grids are planned and operated. Renewable technologies, particularly solar and wind, are essential for decarbonisation, but they also introduce a major operational challenge because their output is intermittent (International Energy Agency, 2023). In practice, renewable generation doesn't always align with demand. Solar and wind output can be high when demand is relatively low, while the grid may experience stress during evening peaks or periods of low wind availability.

Grid-scale Battery Energy Storage Systems (BESS) provide one way to address this mismatch between generation and demand. By charging during periods of surplus generation and discharging when demand is high, batteries act as an essential energy buffer helping stabilise the grid against volatility and supporting higher levels of renewable penetration (Aneke and Wang, 2016). This makes BESS an increasingly important part of future electricity systems.

However, operating BESS effectively isn't straightforward. The economic viability of a battery often depends on electricity price arbitrage, where the system charges when prices are low and discharges when prices are high. A policy that focuses only on short-term profit may cycle the battery aggressively, which can accelerate capacity fade and reduce its State of Health (SoH) (Xu et al., 2018). The long-term value of energy storage depends on control strategies that account for ageing mechanisms alongside revenue.

Compounding this challenge, the battery operates in a shifting market where prices are influenced by a variety of factors creating regimes that may not have been encountered during the training of a control policy. Europe's 2022 gas-price crisis is a concrete instance, moving both the level and the shape of electricity prices well outside any recent training window (International Energy Agency, 2022). This raises a question that goes beyond whether reinforcement learning can learn a good arbitrage policy, can the learned controller remain effective when the market shifts to a regime it has never seen?

B. The Problem

Reinforcement learning (RL) is a suitable framework for battery arbitrage because the problem is sequential by nature. At each timestep, the controller must decide whether to charge, discharge, or remain idle, while accounting for uncertain electricity prices and the long-term impact of degradation. This fits naturally within the Markov Decision Process (MDP) formalism. Recent work has shown that deep RL agents can learn profitable energy-storage arbitrage strategies and can

outperform hand-crafted heuristics (Cao et al., 2020), while Soft Actor-Critic (SAC) is a common choice for continuous-control problems because of its off-policy learning and entropy regularisation (Haarnoja et al., 2018).

However, applying RL to battery arbitrage is complicated by an environment that shifts under the agent. This domain presents three core challenges.

Sample inefficiency. Model-free algorithms such as SAC learn mainly through environment interaction. In this project, one training episode is 14 days of hourly dispatch decisions (336 steps), and training SAC from scratch to convergence takes hundreds of thousands of such steps (Section VI-D) on the order of years of once-a-day real market data if learned model-free and from scratch.

Constraint violations during exploration. Exploration, particularly early in training, can produce physically infeasible actions. Examples include discharge requests that would push the State of Charge (SoC) below its lower operating limit or charge requests that would exceed the upper limit.

Market regime shifts. Battery arbitrage depends on a market process that is itself non-stationary. Electricity prices reflect seasonal demand patterns, renewable availability, weather conditions, and fuel costs, all of which can drift gradually or shift abruptly. A policy calibrated to one regime can encounter very different price spreads and cycling incentives after such a shift.

Together, these challenges motivate a control approach that can learn from few real market days, never explores untrained on the physical system, and adapts to persistent regime shifts at a speed set by days of observation rather than weeks of on-asset fine-tuning.

C. Approach

The objective is fixed from the beginning as maximizing arbitrage profit and minimizing battery degradation, with wear priced at the cost of the health it consumes. To investigate how to pursue that objective from little real data, the dissertation builds a calibrated, fully deterministic market-and-battery simulator as a controlled testbed and studies a family of methods on it. The central method is Dyna where a SAC agent that trains inside a learned model of the environment's dynamics rather than on real interaction alone. Those dynamics split naturally into a market half and a battery half, the dissertation applies Dyna with both halves as the observation, with questions that follow from doing so. Learning the market half turns a season of freely-observed price days into unlimited imagined training days, and the recurring theme throughout the dissertation is the extent to which this approach achieves the performance of model-free arbitrage while relying on substantially less real data. The same structure is then probed from other angles; whether training off-asset keeps exploration safe on the physical battery, whether the framework degrades gracefully under transient shocks, whether a nightly refit-and-replan loop lets it track persistent regime shifts, and whether the battery is better learned than assumed.

This decomposition is motivated by a simple structural observation. A battery operating as a price-taker, the standard modelling assumption for storage small relative to the system it trades in (Sioshansi et al., 2009), interacts with two distinct domains; a deterministic physical system and a stochastic market environment.

We therefore decompose the world model accordingly.

- **The asset physics are known and deterministic, by default.** The State of Charge (SoC) and State of Health (SoH) evolve through well understood electrochemistry, including round-trip efficiency, throughput-based cycling degradation, and calendar ageing.
- **Only the market is stochastic and learned, in the default setup.** Rather than trying to model complex grid-level market drivers, the agent only requires a generative model of 24-hour price vectors. We implement this as an ensemble of hour-autoregressive recurrent networks conditioned on calendar features and historical prices.

This decomposition means only the dynamics of the market half ever needs to adapt to the changing prices, while the battery half remains fixed. A model-free agent gathers at most 24 real transitions per day, no matter how urgently its policy needs updating. The generative model turns those same 24 observed prices into unlimited imagined training days overnight, so the effective sample rate is bounded by compute rather than by the calendar.

D. Research Questions and Contributions

Every investigation in this dissertation shares the premise that a price-taking storage asset's environment decomposes into a battery and an exogenous market price process. A Dyna-style SAC agent trains using imagined transitions validated against held-out data (the market case) or admissibility by construction (the battery case). Because the learned market process is exogenous, the policy's actions can't steer it into self-confirming error, so imagination can run over full 24-step days rather than short branched rollouts. Each investigation below is framed as a question and reported empirically with its finding, positive or negative.

1) **Is it the quality of the world model, rather than the quantity of data or gradient steps, that buys sample efficiency?** The market model, an ensemble of hour-autoregressive networks, is admitted to policy training only if a pre-training gate certifies its imagined days as within the calibration range of the market. Both quantity channels are held open and neither explains the result. More gradient steps do not: a replay-only control on the identical budget, with unlimited extra epochs, finishes far behind. More data does not either: tripling the budget to three years produces an ensemble that fails the gate, because pooling three winters of differing severity smooths away the seasonal extremity one well-covered year preserves. Only the certified model reaches the unlimited-data model-free ceiling (Section VII-A).
2) **Does training in imagination with planner demonstrations produce a deployed policy that requests**

fewer infeasible actions than one trained model-free? No, evaluated on the real asset, the trained Dyna policy requests an SoC-bound-throttled action on 92.8% of steps, statistically indistinguishable from a policy trained entirely model-free (95.1%) or from replay alone (95.2%). The off-asset design doesn't carry over into a deployed policy that violates SoC bounds less often (Section VII-B).

3) **Does the validated world model degrade gracefully under transient shocks it never trained on?** The frozen dyna policy, an unlimited-data model-free policy, and a replay-only control, all trained purely on calm historical data with no shock exposure, are evaluated zero-shot against five unseen shocks on the same shocked days. Dyna clearly outperforms the replay-only control and closely tracks the unlimited-data policy shock for shock, on a real-data budget two orders of magnitude smaller, even edging it under the fuel shock. More calm-regime data doesn't buy shock robustness (Section VII-C).
4) **Can an in-imagination adaptation loop recover from persistent regime shifts?** A nightly refit-and-replan procedure is tested on a deployment where the market shifts partway through. It fails to beat simply freezing the pre-shift policy. The nightly refit's market predictions drift outside the same calibration range used to approve the original model, and never return to it for the rest of the deployment (Section VII-D).
5) **Does a physics-informed battery surrogate need fewer real transitions than a plain MLP to predict the battery's own dynamics?** Holding the market exact, the battery alone is learned from a transition log. On this smooth, low-dimensional map, the physics-informed network shows no robust sample-efficiency crossover over a plain MLP, not even in the small-data regime a PINN is meant to own (Raissi et al., 2019). The finding is negative, and an ablation that refits the network with deliberately wrong physics explains it: the residual carries genuine physical information only at the largest budgets, where breaking it costs about $4.9\times$ the error, and is inert below $N{=}50$, precisely the scarce-data regime that motivates a physics prior in the first place. The zero rate of physically impossible predictions usually claimed for such models is definitional here, imposed by an output clamp that any network can carry, so it isn't evidence for the physics prior either. On this problem the physics-informed approach isn't practical (Section VII-E).

II. Related Work

A. Reinforcement Learning for Battery Arbitrage

Cao et al. (2020) showed deep Q-networks could learn profitable day-ahead bidding that beats linear-programming baselines under price uncertainty; later work applied policy-gradient and off-policy actor-critic methods, with SAC (Haarnoja et al., 2018) emerging as the default for continuous-action energy control. This literature trades short-term profit against long-term asset preservation, but evaluates largely on stationary benchmarks. Model-free adaptation is rate-limited by the market itself, which publishes only 24 real transitions a day, so recovering from a regime shift demands extensive online retraining during which the agent may cycle the asset damagingly. This dissertation takes that adaptation problem as its central question.

B. Model-Based Reinforcement Learning and World Models

Model-based RL improves sample efficiency by learning a dynamics model $\hat{f}(s,a) \to s'$ and generating synthetic transitions from it, canonically through the Dyna architecture (Sutton, 1991); probabilistic-ensemble variants cut the real interactions needed on continuous-control benchmarks by an order of magnitude (Janner et al., 2019; Chua et al., 2018). Their central difficulty is that model error *compounds through the agent's actions* over multi-step rollouts. The price-taker decomposition removes that failure mode outright: the learned process is exogenous, so the policy cannot steer the model into self-confirming error. Latent world models such as Dreamer (Hafner et al., 2020) plan entirely in imagination at the cost of architectural complexity and interpretability, whereas the decomposition here stays inspectable, pairing exact battery physics with a small price model measured against held-out data before it is trusted.

C. Non-Stationarity and Adaptation in Reinforcement Learning

RL under non-stationary dynamics has been studied as worst-case planning against drift (Lecarpentier and Rachelson, 2019), as continual learning balancing plasticity against forgetting (Khetarpal et al., 2022), and surveyed by Padakandla (2021); the concept-drift literature (Gama et al., 2014) contributes the detect-then-adapt pattern followed here. Two features distinguish this setting. The non-stationary component is *exogenous and fully observed*, so adaptation data accrues at a fixed rate for free regardless of the policy. And the asset constraint forbids on-policy re-exploration, so adaptation must run off-asset: refit the market model, retrain in imagination, deploy tomorrow.

D. Physics-Informed Surrogates for Dynamics Models

Physics-informed neural networks (Raissi et al., 2019) penalise violation of a governing law at unlabelled collocation points, so the physics is meant to substitute for data. That promise has been carried into lithium-ion degradation prognosis from limited cell data (Wang et al., 2024), into continuous control (Antonelo et al., 2024), and into the exact architecture used here, since Liu and Wang (2021) train a Dyna-style agent inside a physics-informed dynamics model. Together these motivate inverting this dissertation's default assumption and learning the battery from a log short enough that an operator might plausibly have one. Scope matters for reading the result: the residual is established where the target map is stiff, high-dimensional or sparsely observed, whereas a price-taking battery's step map is smooth, three-dimensional and cheap to label, so Section VII-E measures how far the claim extends rather than whether it ever holds.

III. Synthetic Merit-Order Market Simulator

A. Justification for Synthetic Data

A synthetic, fully observable simulator buys four things real historical data cannot: controlled experimentation, seeded reproducibility, unlimited paired evaluation episodes, and on-demand regime shifts with a known counterfactual. It also makes the evaluation stronger, since the hindsight optimum and the rolling-horizon oracle are exactly computable, so every policy is scored as a fraction of the value actually attainable on the same days. The cost is scope: the claims this dissertation defends are orderings and mechanisms established within the testbed, not absolute profit levels, and none transfer to real markets without validation against measured data (Section VIII-B).

One naming convention follows from this and is used throughout. A *real* day, transition, or year means one drawn from this simulator, as opposed to one generated by a learned model. The contrast is always simulator versus surrogate. No measured market data is used anywhere in this dissertation.

B. Price Formation

Prices form causally from an exogenous driver layer. A calendar and Ornstein Uhlenbeck (OU) weather processes drive a heating-dominated European demand profile, U-shaped over the year, while an OU fuel market drives thermal costs. Against this, a fleet of solar (500 MW, the duck-curve driver), wind, must-run geothermal, reservoir hydro bidding its opportunity cost against reservoir fill, two coal units with minimum-run blocks and nine gas units spanning combined-cycle plants through old peakers submits (price, quantity) offers, and each hour clears as a uniform-price merit-order auction paying all dispatched capacity the marginal price. The thermal fleet is deliberately granular so that units of varying efficiency make the merit order slope, which is what lets the winter premium, the duck-curve spread, negative prices and capped scarcity hours *emerge* from consumption meeting the supply curve rather than being painted onto prices. Shocks perturb the drivers, never prices, so every price movement has an economic mechanism behind it. The full specification is in the submitted source.

C. Calibration

The simulator is held to the statistical calibration contract in Table I, enforced by the test suite over a seeded simulated year. These thresholds were fixed by tuning the fleet and driver parameters until the simulated year showed this structure, not by fitting to measured prices, so no value here is an empirical estimate. They serve as regression bounds, and a failure means the market drifted when a fleet or weather parameter changed.

TABLE I
The calibration contract over a seeded simulated year.

Statistic	Contract
Median price	$\approx$ \$64/MWh
Winter − summer median	$>$ \$6/MWh
Winter − summer mean	$>$ \$15/MWh
Evening peak vs overnight	evening higher
Scarcity hours (clearing at cap)	$< 1.5\%$ of hours
Negative-price hours	rare but present

D. Regime-Shift Scenarios

Regime shifts are expressed through five scenario types: *ColdSnap* (heating-demand surge), *HeatWave* (cooling demand plus thermal derating), *FuelShock* (multiplicative gas-price step), *Drought* (reservoir inflow collapse), and *PlantOutage* (a named unit removed from the fleet). Scenarios perturb *drivers* inside a day window, never prices, and consume no randomness, so the same seed with and without a scenario differs only through the shock's causal consequences. Short windows give the unseen *transient shocks* of the robustness study (Section VI-B); windows extended to the end of the run give the *persistent regime shifts* of the adaptation study (Section VI-C).

IV. Problem Formulation: The Arbitrage MDP

A. Battery Physics

The battery (100 kWh, 50 kW, 90% round-trip efficiency split evenly across the two directions) tracks two states. SoC is expressed as a fraction of the *current degraded* capacity $C_{\text{eff}} = C_{\text{nom}} \cdot \text{SoH}$, and SoH declines through cycling and calendar ageing,

$$\Delta\text{SoH}_{\text{cyc}} = E_{\text{thru}} \cdot \kappa_{\text{cyc}} \cdot \left(1 + s \cdot \tfrac{|P|}{P_{\max}}\right), \tag{1}$$

$$\Delta\text{SoH}_{\text{cal}} = \kappa_{\text{cal}} \cdot \left(0.5 + \overline{\text{SoC}}\right) \cdot \Delta t, \tag{2}$$

where E_{thru} is the step's cell-side energy throughput and κ_{cyc} is calibrated so that 5,000 equivalent full cycles at low power take SoH from 1.0 to the end-of-life floor of 0.8. The stress factor s makes high-power cycling cost more per kWh, standing in for the empirically established super-linear damage of aggressive cycling (Xu et al., 2018). Calendar ageing ticks every hour regardless of action, calibrated to a 15-year rest life at mid SoC, with a full battery ageing three times as fast as an empty one. SoH feeds back into usable capacity, conserving stored energy as capacity shrinks. Requested power is clipped to the power limits and the SoC bounds ($[0.05, 0.95]$), so every executed action is feasible by construction.

B. State, Action, and Episodes

The observation is 32-dimensional: 8 scalars (SoC, SoH, sine/cosine of hour-of-day and of day-of-year, weekend and holiday flags) followed by the *next 24 hourly day-ahead prices*, log-normalized. Prices aren't hidden because day-ahead markets publish them in advance, and the rolling-horizon oracle (Section V-A) shows this window carries 99.9% of the attainable value; weather and fuel states are deliberately *not* observed, influencing reward only through prices already known. A single continuous action $a \in [-1, 1]$ maps to a power command $a \cdot P_{\max}$, positive charging and negative discharging; infeasible commands are clipped by the physics rather than penalised, so degradation, not infeasibility, is the cost channel. Episodes are 14 days (336 hourly steps) with randomised initial SoC and SoH and a burn-in so weather and fuel states

decorrelate. The market operates in MW/MWh and the battery in kW/kWh; settlement converts between them.

C. Reward

The reward monetizes degradation at replacement value rather than weighting it by a free parameter,

$$r_t = \left(\pi_t - \Delta\mathrm{SoH}_t \cdot c_{\mathrm{repl}} \cdot C_{\mathrm{nom}}\right) \cdot \sigma, \tag{3}$$

where π_t is the hour's arbitrage cash flow (buying at the cleared price when charging, selling when discharging, with conversion losses on each side), $\Delta\mathrm{SoH}_t$ is the health lost from both ageing pathways, $c_{\mathrm{repl}} = \$250$/kWh is the replacement cost, and σ a fixed reward scale. Wear is priced at what it costs to buy back rather than weighted by a free parameter, so every policy optimises the same dollars. Calendar ageing makes doing nothing strictly costly, anchoring the low end of the benchmark ladder.

V. Methodology

A. Oracles and Baselines

Every result is compared against four reference policies.

- **Hindsight optimum.** The best schedule possible, found by dynamic programming over a discretised grid of State-of-Charge values, using the actual prices for the whole episode. It's a valid upper bound because the battery is a price-taker, so its trading can't change the prices it plans against. Perfect-foresight dynamic programming under exactly this assumption is the established way the arbitrage value of storage is bounded (Sioshansi et al., 2009). This defines 100% of the value any policy could attain.
- **Rolling-horizon oracle.** The same dynamic program, but limited to what the agent actually knows at each step (the next 24 known prices), and re-run every hour. It reaches 99.9% of the hindsight optimum. This tells us the 24-hour window already contains almost all the information needed, so any gap left by a learned policy is a failure to learn, not a lack of information. A test pins its cost model to the real battery physics, so "optimal" really means optimal against the true dynamics. In the adaptation experiment it also serves as the *instant-adaptation baseline*, since it replans on each day's observed prices and so adapts immediately by design.
- **Heuristic.** A simple price-threshold rule that charges when the price is in a low percentile of the known window and discharges when it's in a high one. It stands in for a trusted human operator.
- **Idle.** This arm does nothing at all. It still loses money because the battery ages over time (calendar ageing), so it marks the floor.

Most of the arbitrage value comes from a few scarcity weeks, so profit varies a lot from one market week to the next. To keep comparisons fair, each one uses at least 20 paired episodes that share the same random seed, and results are reported as a percentage of the hindsight optimum rather than in raw dollars.

B. The Learned Market Model

The world model is a generative model of a single market day. It produces 24 hourly prices (log-normalized) from a 7-dimensional context made up of four calendar features (position in the year, weekend, holiday) and three summary statistics of the previous day's prices (log-normalized mean, minimum, and maximum). Generation is *hour-autoregressive*, which means the model produces the day one hour at a time. The context sets the initial hidden state of a gated recurrent unit (GRU, Cho et al., 2014), which then steps through the 24 hours; each hour's price is drawn from a Gaussian that depends on the hours already generated. To capture the model's own uncertainty, we train an ensemble of $K = 5$ networks, each fit on a bootstrap resample of the observed days, following the standard practice of treating disagreement across independently fitted networks as a predictive-uncertainty estimate (Lakshminarayanan et al., 2017; Osband et al., 2018). Each imagined episode picks one ensemble member and generates day after day, feeding each generated day back as the context for the next.

We also record two architectures that didn't work. A feedforward model that generates all 24 hours in one shot as independent Gaussians (and a low-rank variant that allows limited hour-to-hour correlation) fails the validation gate. Because these models treat the 24 hours as independent, they can't represent the fact that scarcity comes in blocks of several consecutive expensive hours. To reproduce those hours at all, they have to give every hour a fat tail, which makes the daily price spread far too wide. The hour-autoregressive model avoids this. Once it effectively "decides" that a day is a scarcity day, it stays on that path for the rest of the day.

C. The Validation Gate

Before it's used for any policy training, the fitted ensemble must pass a statistical gate. We generate synthetic years from the model and score them on the same calibration statistics the simulator itself is tested against, so the learned model has to meet the same standard as the simulated market it was fitted to. For checking price spread, the reference is a held-out real year generated from a seed that isn't part of the training data.

The criteria cover the market properties that arbitrage value depends on. The synthetic median price must fall within the calibrated band. The winter price premium must show up in both the median and the mean, and evening prices must be higher than overnight prices. The mean daily spread must be within 30% of the held-out real year, which catches both models that invent too much spread and models that smooth it away. Finally, scarcity hours must stay below 1.5% of all hours, and negative-price hours must occur but stay below 15% of all hours, so a model can't pass while getting wrong the exact hours where most of the value is.

A model that fails any criterion isn't used and the Dyna training script refuses to run unless it's given a model that has passed the gate. In standard model-based RL, trust is usually controlled by limiting how far the model is rolled out, because prediction errors compound as the policy acts. Here they can't, because the policy can't affect prices, so the only question is

whether the generated days have the right distribution, and that can be checked *before* any RL compute is spent.

The gate is also what makes model quality a measurable variable rather than an assumption. Data quantity and gradient count are both trivially observable, so an experiment can hold them fixed by construction; the calibration of the imagined days is not, and without a statistic for it the sample-efficiency question reduces to comparing arms that differ in ways nobody has quantified. Two design properties matter for using it this way. It costs a few seconds of sampling rather than a policy training run, so it can screen candidate models before any of them are trained. And it is a far lower-variance statistic than downstream profit, which in this market is dominated by a handful of scarcity weeks (Section V-A) and would make a noisy model-selection signal even though exact off-asset policy evaluation is available here.

D. Dyna Training with Planner Demonstrations

The Dyna agent is a standard SAC agent trained inside the learned market, with one addition. Before training begins, we fill part of the replay buffer with transitions from the rolling-horizon planner acting on the D stored *real* days (replayed exactly through the environment). These planner demonstrations act like the logs of a trusted existing controller. Seeding an off-policy replay buffer with demonstrations is established practice for cutting the cost of early exploration (Hester et al., 2018; Vecerik et al., 2017) but what differs here is the source, since the demonstrator is an exact planner rather than a human operator. This is to prevent the battery from untrained random actions. They put real-market transitions and sensible behaviour into the buffer without the policy ever having to act on the real system before it's trained. From then on, all exploration happens in imagination. Checkpoints are chosen using only each arm's own data source, the real simulator is used only for the final held-out evaluation.

E. The Adaptation Loop

The agent is deployed in a streaming setup where it sees one real market day at a time and an adaptation step runs at the end of each day. Each night, the adaptive agent does the following.

1) **Observes and stores** the day's 24 published prices, adding them to its history. This data arrives for free, no matter what the policy did.
2) **Detects** how surprising the day was, by computing the negative log-likelihood (NLL) of the observed day under the ensemble, the detect-then-adapt pattern of the concept-drift literature (Gama et al., 2014) with likelihood under the current model as the drift statistic. A regime shift surprises every ensemble member at once, so this per-day NLL is a clearer signal than measuring how much the members disagree. The NLL over the whole deployment is itself one of the reported results. (The main version adapts every night regardless; a version that only adapts when the NLL is high is a compute-saving refinement.)
3) **Refits the market model.** It fine-tunes each ensemble member, starting from its current weights, on the most recent 90 days of observed prices plus a freshly-resampled set of 30 pre-deployment "anchor" days that guard against forgetting, the plasticity-stability trade-off that continual learning has to manage (Khetarpal et al., 2022). It's deliberately *not* retrained from scratch, because much of the old structure (the daily shape, calendar effects) still holds and should be kept. Each member resamples its data (bootstrap) so the ensemble stays diverse.
4) **Refreshes demonstrations.** It runs the rolling-horizon planner on the last 7 *real* (post-shift) days and adds those transitions to a small replay buffer, so the policy update is grounded in real shifted behaviour and not only in imagination.
5) **Fine-tunes the policy in imagination.** It continues SAC training from the current weights for a fixed number of steps inside the updated learned market. The imagination environment refers to the ensemble directly, so every generated day already reflects the newly refitted model.
6) **Deploys the next day.** All of the steps above run off the physical asset, overnight.

The model-free alternative, which fine-tunes SAC directly on the real price stream, is limited to just 24 new transitions per day. The gap between these two approaches is exactly what the experiment measures.

VI. Experimental Setup

A. Experiment 1: Sample Efficiency under a Real-Data Budget

Each arm is given exactly D days of real market history and nothing more.

- **online.** Standard SAC trained on the real simulator and stopped after $24 \cdot D$ steps (the number of hours in D days). This shows what plain model-free learning can extract from the budget.
- **replay.** SAC trained on the same D stored days replayed exactly, but with unlimited gradient updates. This is the control that separates the effect of "more gradient updates" from the effect of "having a generative model".
- **dyna.** SAC trained in the learned market that was fitted to the same D days (when the gate allows it), with the replay buffer seeded by planner demonstrations on those days (Section V-D).

All arms are evaluated the same way, over 20 paired 14-day episodes on the real simulator, and reported as a percentage of the hindsight optimum. Varying D shows how both policy performance and gate pass/fail depend on the amount of data. An *unlimited-data* SAC reference, trained far beyond any budget, marks the model-free ceiling.

B. Experiment 2: Zero-Shot Robustness to Transient Shocks

The budget-trained policies are frozen and then tested under the five scenario types from Section III-D, applied as short shock windows inside the evaluation episodes. These are regimes the policies never saw during training. Because the

scenarios consume no randomness, each shocked episode can be paired with a calm episode from the same seed, which isolates the effect of the shock exactly. The question isn't whether performance drops (it drops for everything) but how gracefully each policy degrades compared with the oracle on the same shocked days.

C. Experiment 3: Adaptation to Persistent Regime Shifts

Each deployment is one continuous 120-day market run. The first 30 days are calm and establish each arm's baseline then a persistent shift begins at day $T = 30$ and lasts until the end. There are three shifts, chosen to span a simple level change versus a structural change.

- **fuel-step.** The gas price doubles ($\times 2$) permanently. This is a pure change in price *level*, which the 24-hour observation window already conveys quite well, so the frozen policies should cope best here.
- **capacity-loss.** The cheap 100 MW baseload coal unit (with its minimum-run block) is permanently lost. This is a *structural* change to the merit order that alters *when* scarcity and negative prices occur, not just how high prices are.
- **cold-regime.** A lasting increase in heating demand, which shifts both the level of demand and its seasonal pattern.

The five arms compared in each deployment are:

- **frozen-dyna** and **frozen-ideal.** The dyna and unlimited-data policies, never updated during the deployment.
- **online-ft.** The unlimited-data policy fine-tuning directly on the real price stream, which provides only 24 new transitions per day.
- **adaptive-dyna.** The full nightly adaptation loop of Section V-E.
- **heuristic.** The fixed price-threshold rule of Section V-A.

All arms, together with the rolling-horizon **oracle**, run on the same paired seeds. Prices are exogenous, so any two arms sharing a seed see exactly the same market.

All metrics are normalized by the oracle on the same seed. Because the oracle replans on each day's observed prices, it adapts instantly by construction, which makes it a fair "instant adaptation" ceiling.

- **capture**(t), the policy's net profit over a trailing 7-day window divided by the oracle's, i.e. the fraction of the instantly-adapted optimum the policy keeps;
- **recovery lag**, how many days after T it takes for capture to return to 90% of the arm's *own* pre-shift level;
- **cumulative regret**, the total post-shift shortfall against the oracle, in dollars;
- **detection trace**, the adaptive arm's per-day NLL, before and after each nightly fine-tune, showing whether and when the model noticed the shift.

D. Hyperparameters, Seeds, and Reporting

All SAC arms use Stable-Baselines3 (Raffin et al., 2021) defaults, except warmup, which scales with the step budget $(\min(2{,}000, \max(T/10, 100)))$ so the small online budgets aren't burned entirely on random actions. Replay and dyna train for 500,000 steps across 4 parallel envs, checkpointed by periodic 5-episode evals on each arm's own data source (Section V-D). The market ensemble (Section V-B) is 5 GRUs, hidden width 128, 2,000 epochs. The nightly loop (Section V-E) fine-tunes for 25 epochs and trains the policy for 10,000 imagined steps; online-ft instead takes 240 real gradient steps a night. Seed ranges never overlap: training at seed 0, held-out eval from seed 1,000,000, adaptation deployments from seed 7,000,000.

Dollar amounts scale with the simulator's price cap, because most arbitrage value comes from capturing scarcity hours. We measured this sensitivity rather than assuming it. The hindsight optimum is about $+\$90$ per 14-day episode at a \$3000/MWh cap and $+\$28$ at \$1000, and at \$200 the heuristic falls below idle because spread arbitrage alone no longer covers the cost of cycling the battery. We fix the cap at \$1000/MWh for all experiments, chosen so the benchmark ladder stays strictly ordered and evaluation variance stays manageable. Every headline number is reported as a fraction of the oracle or hindsight optimum on the same seeds, which cancels out this level scaling. The learned-policy comparisons are all run at this single cap; Section VIII-B states what is and isn't claimed as a result.

VII. Results and Discussion

A. Does Model Quality, Rather Than Data Quantity, Buy Sample Efficiency?

Unless stated otherwise, quoted percentages are of the hindsight optimum on the real simulator (20 paired 14-day episodes for the budget and robustness experiments), and of the same-seed rolling-horizon oracle (10 paired 120-day deployments) for the adaptation experiment. Because these are ratios of pooled sums rather than means of per-episode ratios, their error margins are leave-one-out jackknife standard deviations over the resampling unit named in each caption. Quantities that are not pooled ratios, namely regret in dollars and surrogate error across fit seeds, carry plain standard deviations instead.

Table II sweeps $D \in \{90, 365, 1095\}$ real days for the hour-autoregressive ensemble, and only $D = 365$ passes every criterion. At $D = 90$ it fails in the expected direction, since barely a season means the invented daily spread comes out more than double the real year's and the winter/evening patterns flip sign entirely. At $D = 1095$ it fails differently. Pooling three winters of varying severity smooths away exactly the seasonal extremity one well-covered year preserves. So the gate isn't rewarding "more data" as such, it's picking out one specific window, and neither more nor less data was assumed safe going in.

At $D = 365$, the only budget that passes, dyna hits $71.0 \pm 14.8\%$ of hindsight (best checkpoint). Replay, on identical data and gradient budgets, reaches $38.8 \pm 32.3\%$ at the same budget (and at most 54.1% at any budget), behind dyna by more than either margin, so it's the generative model doing the work, not extra gradient steps. The unlimited-data reference hits

TABLE II

VALIDATION-GATE OUTCOME ACROSS REAL-DATA BUDGETS D (DAYS), HOUR-AUTOREGRESSIVE $K = 5$ ENSEMBLE. SYNTHETIC VALUE SHOWN PER CELL; ✓/× MARKS WHETHER THE CRITERION PASSES (SECTION V-C). EVERY ROW IS JUDGED AGAINST AN ABSOLUTE THRESHOLD EXCEPT MEAN DAILY SPREAD, WHICH IS JUDGED RELATIVE TO THE REAL COLUMN (WITHIN 30%).

Metric	Real	D=90	D=365	D=1095
Median ($/MWh)	67.8	63.1 ✓	62.4 ✓	61.6 ✓
Winter median gap	8.1	11.6 ✓	12.2 ✓	10.3 ✓
Winter mean gap	27.5	−44.3 ×	17.1 ✓	10.4 ×
Evening − night	51.0	−19.7 ×	31.1 ✓	32.5 ✓
Mean daily spread	84.5	185.8 ×	91.1 ✓	72.7 ✓
Cap share	0.005	0.007 ✓	0.002 ✓	0.002 ✓
Negative share	0.007	0.007 ✓	0.002 ✓	0.001 ✓
Gate		**FAIL**	**PASS**	**FAIL**

$72.5 \pm 13.1\%$; the gap between it and dyna is well inside one jackknife standard deviation, so one year of real data through the learned market recovers the model-free ceiling, statistically, on two orders of magnitude less real history (Figure 1).

Both quantity explanations are therefore closed off from opposite ends. Extra gradient steps are ruled out by replay, which sees the same year and takes as many epochs as it wants. Extra data is ruled out by the gate sweep, where three years of history produce a worse-calibrated ensemble than one (Table II) and so never reach policy training at all. What separates the arms is the calibration of the days the agent trains on, not how many days produced them or how often it revisits them. One control is missing from this argument and should be named. Because a failing ensemble is never saved, no dyna arm was trained inside a model the gate rejected, so the sweep shows that certified models train good policies without showing that rejected ones train bad policies. Supplying that arm is the cheapest available strengthening of this result (Section VIII-C). Dyna is a single point rather than a curve, since the gate refused the other two budgets; online and replay both improve with more data but stay well below the oracle across the whole range.

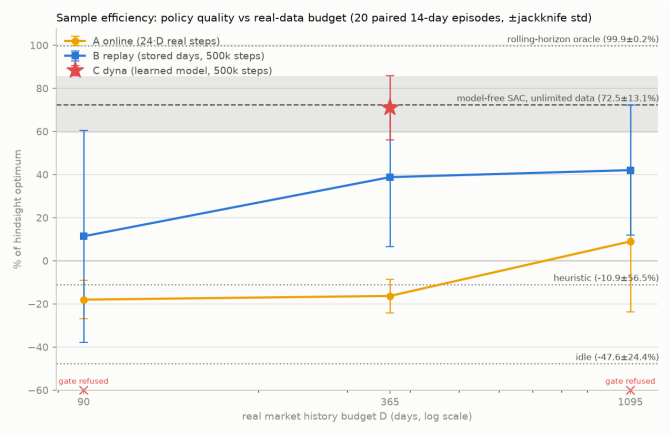

Fig. 1. Policy quality (% of hindsight optimum) versus real-data budget D, 20 paired 14-day episodes per point. Dyna exists only at $D = 365$, the only budget its market model passed the validation gate at.

B. On-Asset Feasibility: Training-Time Exploration versus Deployment

Only the rolling-horizon planner generates real transitions during Dyna training, and it solves a dynamic program over an SoC grid confined to $[\text{soc\_min}, \text{soc\_max}]$, so an infeasible move is never in its search space to begin with. SAC's own exploration still tries infeasible actions constantly, since that's how RL exploration works, but only inside the imagined market against the exact battery physics, never against the real asset. So training doesn't avoid violations, it just keeps them off the physical battery. Table III (top block) confirms this over 20 paired 14-day episodes. A model-free agent exploring on the asset requests something infeasible on $35.8 \pm 3.3\%$ of its steps, mostly during warmup; even the heuristic hits $29.4 \pm 0.1\%$ while the planner hits zero.

But that guarantee doesn't survive deployment. Table III's bottom block shows all three trained checkpoints requesting an SoC-throttled action on over 90% of steps, and dyna-365 ($92.8 \pm 6.7\%$) is statistically indistinguishable from replay-365 ($95.2 \pm 5.8\%$) and the fully model-free unlimited-data policy ($95.1 \pm 4.9\%$).

Why? The natural suspicion is careless training, but a follow-up measurement points elsewhere. We recorded, for the same episodes, the share of steps each deployed policy spends with its SoC already within 0.1 of an operating bound (the table's final column), and it tracks the request rate almost exactly: $95.6 \pm 4.4\%$ for dyna-365, and indistinguishably for the other two arms. A trained policy learns to sit near full or near empty, since that's what buying low and selling high looks like; once parked at a bound, almost no room is left, so any further push in the same direction gets throttled. The near-coincidence of the two columns says the requests are overwhelmingly of this parked kind, pushes whose executed effect is a no-op, rather than deep violations mid-range. That is also why the training method stops mattering: the cause is ordinary converged-policy behaviour, not careless training, so there's no reason the safer, off-asset pipeline would avoid it any more than the other two arms do. So RQ2's answer splits cleanly. During data collection the answer is yes, and it's worth roughly 3,100 avoided infeasible requests per training year (Table III, top block). After deployment the answer is no, but the follow-up shows the residual requests are bound-parking pressure rather than dangerous excursions; what a real battery-management system would make of ∼8,000 such refused pushes a year, benign or not, is a deployment question this simulator can't settle (Section VIII-B).

C. Does the Model Degrade Gracefully Under Shocks?

Yes it does but not perfectly. Frozen dyna-365 keeps $58.1 \pm 6.4\%$ to $90.0 \pm 2.6\%$ of the oracle's capture depending on the shock (Table IV), tracking the unlimited-data policy shock for shock on a real-data budget two orders of magnitude smaller; no shock separates the two arms by more than their jackknife margins (under the fuel shock dyna's point estimate is even ahead, $58.1 \pm 6.4\%$ vs. $56.4 \pm 6.4\%$, though the margin covers it). Fuel-shock is hardest for every arm, since it perturbs the

TABLE III

INFEASIBLE ON-ASSET ACTION REQUESTS (AN SOC BOUND THROTTLING MORE THAN 1% OF THE REQUESTED ENERGY), 20 PAIRED 14-DAY EPISODES ON THE REAL MARKET, MEAN ± STD OVER EPISODES. THE MIDDLE COLUMN EXTRAPOLATES TO A 365-DAY BUDGET OF ON-ASSET STEPS AT THE MEAN RATE. THE TOP BLOCK LISTS BEHAVIOURS THAT WOULD SOURCE REAL TRANSITIONS DURING TRAINING, THE BOTTOM BLOCK THE FULLY TRAINED POLICIES AS RUN AT EVALUATION; FOR THOSE, THE FINAL COLUMN GIVES THE SHARE OF STEPS SPENT PARKED WITHIN 0.1 OF AN SOC BOUND, THE FOLLOW-UP MEASUREMENT THAT ATTRIBUTES THE REQUEST RATE.

On-asset behaviour	Infeasible-request rate	Per 365-day budget	Near-bound share
Random (model-free warmup exploration)	$35.8 \pm 3.3\%$	≈3,100	n/a
Heuristic (fixed rule)	$29.4 \pm 0.1\%$	≈2,600	n/a
Rolling-horizon planner (Dyna data source)	$0.0 \pm 0.0\%$	0	n/a
Idle	$0.0 \pm 0.0\%$	0	n/a
Trained dyna-365 (deployed, deterministic)	$92.8 \pm 6.7\%$	≈8,129	$95.6 \pm 4.4\%$
Trained replay-365 (deployed, deterministic)	$95.2 \pm 5.8\%$	≈8,343	$96.2 \pm 4.3\%$
Trained unlimited-data (deployed, deterministic)	$95.1 \pm 4.9\%$	≈8,334	$96.5 \pm 3.4\%$

price process in a way the 24-hour window conveys less directly. Cold-snap and plant-outage, by contrast, actually leave dyna-365 and unlimited-data *above* their calm capture, because those shocks mostly raise a price level the window already shows. Replay is the weakest arm throughout, and by margins its wide uncertainty can't excuse (its best shock, $47.1 \pm 14.8\%$, sits below dyna's worst), which reads as a general policy-quality gap rather than a shock-specific one. The calm-trained controller is therefore a floor rather than a solution: it doesn't need shock exposure to avoid disaster, but the fuel-shock column shows how much a level-shifting regime still costs.

D. Can the Adaptation Loop Recover from Persistent Shifts?

No, across all three shifts and 10 paired deployments each, adaptive-dyna never beats simply freezing the pre-shift policy (Table V). Both arms run identical deployments, so the paired difference is the test: frozen-dyna's post-shift capture exceeds adaptive-dyna's by 16.3 ± 3.5, 11.0 ± 1.4 and 8.9 ± 1.7 percentage points on fuel-step, capacity-loss and cold-regime. Adaptive-dyna is behind before the shift even begins, by 22.0 ± 4.1 points on fuel-step. Table VI splits that deficit in two: an acute startup transient over the first fortnight, while the ensemble fine-tunes on a data-starved history, and a smaller steady-state penalty that never closes.

None of the pre-registered hypotheses (Section VI-C) survive. Recovery lag orders differently on every shift, and is measured against an arm's *own* pre-shift level, so adaptive-dyna "recovering" only means returning to an already-worse baseline. Frozen-dyna copes worst with the pure level shift predicted to suit it best (53.4% on fuel-step) and best with the structural shift predicted to be hardest (88.7% on capacity-loss). The NLL trace spikes as often before the shift as after, so it's too noisy to trigger anything on its own.

1) Where the shortfall comes from: The two suspects are the nightly market refit and the policy fine-tune. To separate them we replayed the refit offline against the fuel-step deployment, using the loop's exact configuration (90-day window, 30 anchor days, 25 epochs), and re-scored the ensemble after every night against the same gate that admitted the original 365-day fit.

The refit leaves the gate on its third night and never returns across the remaining 116 (`scripts/measure_ensemble_drift.py` reproduces the full trace). The original ensemble's mean daily spread (83.6) sits inside the required band (59.2–109.9); by night three it's at 141.3. The shift doesn't begin until day 30, so the model has already drifted out of calibration 27 days before the market changes at all. Nightly refitting degrades the ensemble on stationary data, which is also why the adaptive arm trails the frozen one *before* the shift (Table VI), when there's nothing to adapt to.

That timing also rules out the obvious repair. Re-gating the nightly refit, freezing the ensemble on any night it fails, would freeze it on night three; and an adaptive arm whose model stops updating on night three is frozen-dyna, the arm that already wins. The gate can prevent this failure but can't convert it into successful adaptation. Incremental nightly fine-tuning on a rolling window inflates the model's price spreads whether or not the market has moved, and SAC then trains inside a model whose swings are systematically larger than the real market's, so the policy is shaped to exploit spreads that don't exist. The earliest refits have the least data and are the most miscalibrated, which is the startup transient.

E. Does a Physics Prior Need Fewer Transitions?

No, and not even in the small-N regime a PINN is meant to own (Section II-D, Table VII). Holding the market exact, an MLP and a physics-informed network both learn $(\text{soc}, \text{soh}, \text{power fraction}) \mapsto (\Delta\text{soc}, \Delta\text{soh}, \text{grid energy})$ from a shrinking budget of N transitions, over ten fit seeds. The soft residual beats the MLP only at $N{=}500$ (0.014 ± 0.002 vs. 0.018 ± 0.003), and its apparent edge at $N{=}200$ sits inside one standard deviation. Below that the MLP wins outright and by widening margins, the battery map being smooth enough that an unconstrained network interpolates it from a handful of points. Under a narrow training distribution the MLP wins at every budget.

The unconstrained MLP does predict impossible transitions, up to $17.9 \pm 3.3\%$ of test points at $N{=}10$, but the output clamp that rules these out isn't free either: `pinn-hard` is the `mlp` arm with clamped outputs, and it's the less accurate of the two at every budget.

TABLE IV

ZERO-SHOT CAPTURE (% OF THE SAME-REGIME ROLLING-HORIZON ORACLE, POOLED RATIO ± LEAVE-ONE-EPISODE-OUT JACKKNIFE STD) UNDER THE CALM CONTROL AND FIVE TRANSIENT SHOCKS, 20 PAIRED 14-DAY EPISODES, SEED0= 1,000,000 (`SCRIPTS/MEASURE_SCENARIO_ROBUSTNESS.PY`). NONE OF THE THREE POLICIES SAW ANY SHOCK DURING TRAINING.

Policy	Calm	Cold-snap	Heat-wave	Fuel-shock	Drought	Plant-outage
dyna-365	71.1 ± 14.8	90.0 ± 2.6	83.1 ± 8.2	58.1 ± 6.4	72.9 ± 18.2	87.3 ± 3.5
replay-365	38.9 ± 32.4	43.7 ± 6.9	33.1 ± 18.2	19.1 ± 9.7	43.0 ± 43.4	47.1 ± 14.8
unlimited-data	72.6 ± 13.1	90.8 ± 1.9	83.0 ± 7.5	56.4 ± 6.4	72.1 ± 18.3	89.4 ± 2.3

TABLE V

ADAPTATION RESULTS, ALL THREE SHIFTS, 10 PAIRED DEPLOYMENTS EACH. CAPTURE IS POOLED 7-DAY-WINDOWED NET PROFIT OVER THE SAME-WINDOW ORACLE NET PROFIT, ± LEAVE-ONE-DEPLOYMENT-OUT JACKKNIFE STD; REGRET IS MEAN ± STD OVER DEPLOYMENTS; RECOVERY LAG IS DAYS AFTER T UNTIL AN ARM REGAINS 90% OF ITS *own* PRE-SHIFT CAPTURE (NEVER = NOT WITHIN THE 90-DAY POST-SHIFT HORIZON), COMPUTED ON THE POOLED CURVE (`SCRIPTS/ADAPTATION_STATS.PY`).

Shift	Arm	Pre capture (%)	Post capture (%)	Recovery lag	Regret ($)
fuel-step	adaptive-dyna	66.5 ± 10.3	37.1 ± 7.2	19d	231 ± 85
fuel-step	frozen-dyna	88.5 ± 8.0	53.4 ± 8.1	20d	172 ± 47
fuel-step	frozen-ideal	91.8 ± 5.0	41.6 ± 8.7	never	214 ± 74
fuel-step	online-ft	85.0 ± 5.8	12.3 ± 8.3	never	328 ± 193
fuel-step	heuristic	54.7 ± 25.3	27.1 ± 9.5	14d	282 ± 111
capacity-loss	adaptive-dyna	63.4 ± 14.3	77.7 ± 4.6	1d	218 ± 119
capacity-loss	frozen-dyna	90.2 ± 7.2	88.7 ± 3.5	7d	110 ± 54
capacity-loss	frozen-ideal	90.5 ± 5.1	91.1 ± 2.1	1d	90 ± 55
capacity-loss	online-ft	78.2 ± 6.7	66.0 ± 6.6	9d	348 ± 301
capacity-loss	heuristic	54.5 ± 25.1	55.2 ± 6.2	3d	467 ± 309
cold-regime	adaptive-dyna	71.2 ± 17.4	74.0 ± 3.1	8d	857 ± 858
cold-regime	frozen-dyna	90.3 ± 7.2	82.9 ± 4.2	1d	546 ± 695
cold-regime	frozen-ideal	90.6 ± 5.1	87.6 ± 2.0	1d	394 ± 443
cold-regime	online-ft	80.4 ± 7.7	83.8 ± 1.6	1d	538 ± 412
cold-regime	heuristic	54.5 ± 25.1	69.8 ± 1.7	1d	982 ± 851

TABLE VI

ADAPTIVE-DYNA'S SETTLED-CALM CAPTURE (7-DAY WINDOWS FULLY INSIDE DAYS 20–29, AFTER THE STARTUP TRANSIENT) VERSUS FROZEN-DYNA, AND BOTH ARMS' SETTLED POST-SHIFT CAPTURE (WINDOWS FULLY INSIDE DAYS 37–119, AFTER THE SHIFT'S OWN SETTLING WINDOW). POOLED CAPTURE ± JACKKNIFE STD, IN %.

Shift	Adapt. (calm)	Frozen (calm)	Adapt. (post)	Frozen (post)
fuel-step	83.7 ± 9.0	94.0 ± 5.1	40.7 ± 7.9	55.6 ± 8.8
capacity-loss	80.3 ± 11.0	93.4 ± 3.8	78.5 ± 3.4	89.7 ± 2.4
cold-regime	81.8 ± 9.9	93.8 ± 3.7	75.1 ± 3.1	82.3 ± 4.5

TABLE VII

HELD-OUT POOLED RELATIVE MAE (ABSOLUTE ERROR OVER TARGET STD, POOLED ACROSS THE THREE TARGETS), UNIFORM COVERAGE, MEAN ± STD OVER TEN FIT SEEDS. **BOLD** IS BEST POINT ESTIMATE AT EACH BUDGET. `PINN-HARD` IS THE SAME TRUNK AS `MLP` WITH THE OUTPUTS CLAMPED TO PHYSICALLY POSSIBLE VALUES AND NO RESIDUAL, SO THE `MLP` TO `PINN-HARD` STEP ISOLATES THE CLAMP AND THE `PINN-HARD` TO `PINN-SOFT` STEP ISOLATES THE PHYSICS RESIDUAL.

N	mlp	pinn-hard	pinn-soft
500	0.018 ± 0.003	0.020 ± 0.005	$\mathbf{0.014 \pm 0.002}$
200	0.034 ± 0.003	0.041 ± 0.005	$\mathbf{0.032 \pm 0.003}$
100	$\mathbf{0.043 \pm 0.003}$	0.058 ± 0.006	0.065 ± 0.006
50	$\mathbf{0.076 \pm 0.006}$	0.128 ± 0.010	0.170 ± 0.020
25	$\mathbf{0.147 \pm 0.009}$	0.151 ± 0.014	0.187 ± 0.016
15	$\mathbf{0.358 \pm 0.014}$	0.498 ± 0.043	0.521 ± 0.011
10	$\mathbf{0.454 \pm 0.029}$	0.648 ± 0.033	0.504 ± 0.013

1) Wrong-constraints ablation: To be thorough about *why* the prior doesn't pay, we asked whether a PINN given the wrong physics could still beat the MLP. If it could, the residual was never carrying physical information and its occasional win was generic regularization. We refit the same tuned soft-residual PINN three times with a broken law: efficiency frozen at 0.60 instead of the true 0.949, nameplate capacity halved in the energy identity, and the calendar-ageing law flipped so an empty battery ages fastest. A control freezes efficiency at its *true* value, separating "frozen" from "wrong".

It can't (Table VIII). At N=500 wrong efficiency (0.070 ± 0.004) and wrong capacity (0.071 ± 0.002) cost about $4.9\times$ the correct residual's error and land about $3.9\times$ worse than the MLP, while the control (0.014 ± 0.002) matches the correct residual exactly. So where the residual helps at all, it helps because the physics is right. But the penalty for wrong physics shrinks steadily as data thins ($4.9\times$, $2.7\times$, $1.6\times$, $1.1\times$ at $N = 500, 200, 100, 50$) and disappears below N=50, where every wrong variant sits inside the seed spread of the correct one.

The residual is therefore inert exactly where a PINN is supposed to help. It only bites once there's enough data to identify its physical parameters, which is the regime where the plain MLP was already good enough. Taken with the accuracy results, the verdict on this problem is simply negative: the physics prior isn't practical here. It loses on accuracy at every

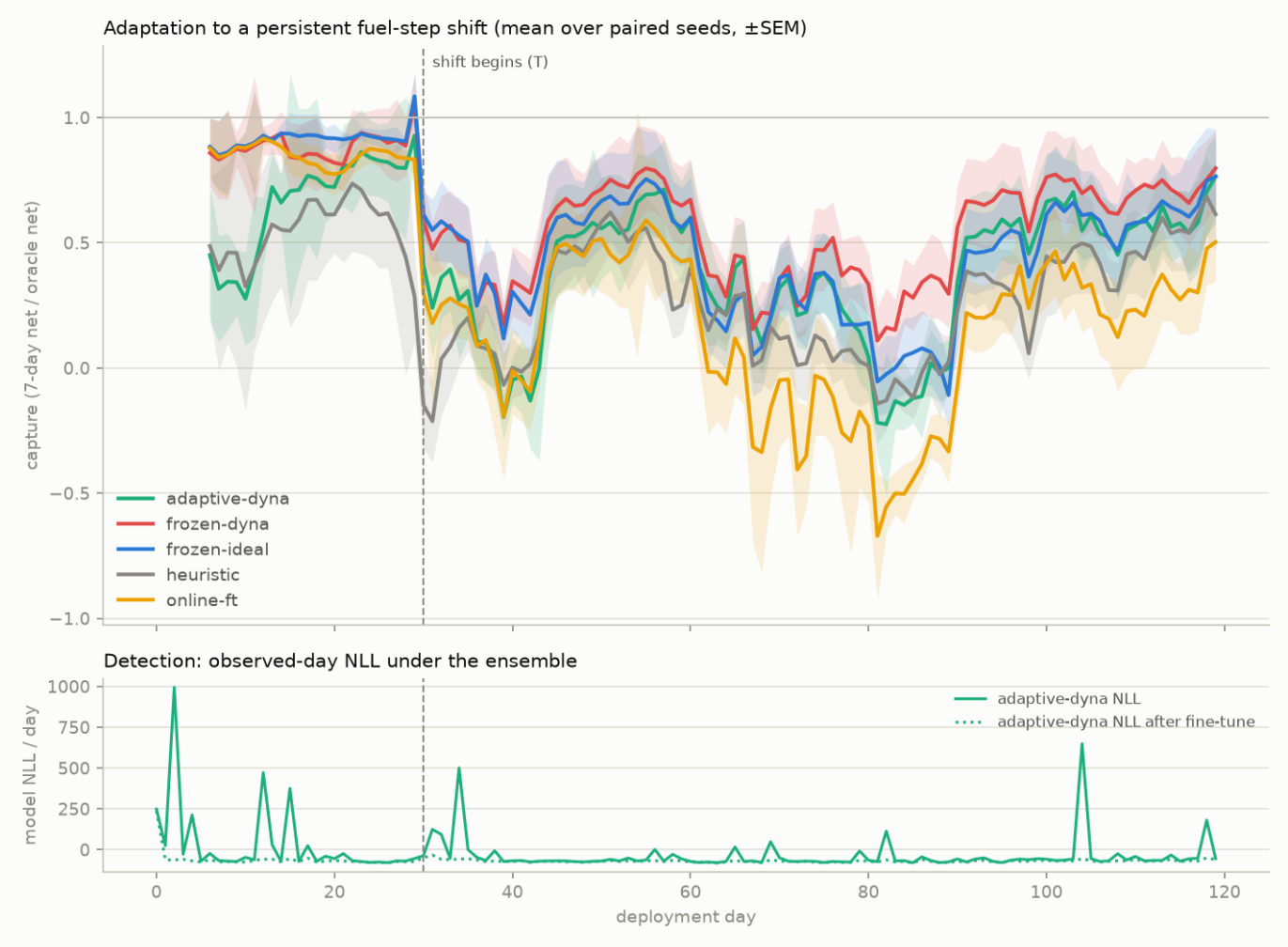

Fig. 2. Capture curves and NLL detection trace, fuel-step shift, mean over 10 paired seeds (±SEM).

budget an operator would realistically have, its residual carries no information at those budgets, and its one apparent guarantee is an output clamp restated, available to any network without physics. Section VIII-B notes what would have to change for the conclusion to differ.

VIII. Conclusion and Future Work

A. Summary of Findings

The results show that watching the market for a single year can be enough, on the one budget of the three tried whose fitted model passed the gate. Feed that year into a generative model of daily prices, check the model against a held-out year from the same simulator, and train the agent inside it, and the resulting policy is as good as one trained on unlimited market experience ($71.0 \pm 14.8\%$ of the hindsight optimum against $72.5 \pm 13.1\%$). The comparison that matters is the replay control, which sees exactly the same year and gets as many gradient updates as it wants, and still lands 32 points behind. So the gain doesn't come from training harder on the same data. It comes from having a model good enough to invent new days to train on. And "good enough" isn't a judgement call here: the gate decides it against fixed criteria, before any training compute is spent. Nor is more of anything a substitute for it. Three years of history make a worse-calibrated model than one and never reach training, and the replay arm shows that grinding more gradient steps through the same year does not close the gap. Quality of the training distribution is the operative variable, and the gate is what measures it. Two boundaries on that claim are worth keeping in view. The gate scores the model against a held-out year from the same simulator, on thresholds set by tuning that simulator rather than by fitting measured prices (Section III-C), so a pass certifies internal calibration, not realism. And no policy was ever trained inside a rejected model, so the sweep establishes that certified models train good policies without establishing the converse (Section VIII-C).

Beyond that, the dissertation maps where the same decomposition stops paying. Over hours there's nothing to fix, since the 24-hour price window already carries almost all the value and the rolling-horizon oracle reaches 99.9% of hindsight. Over

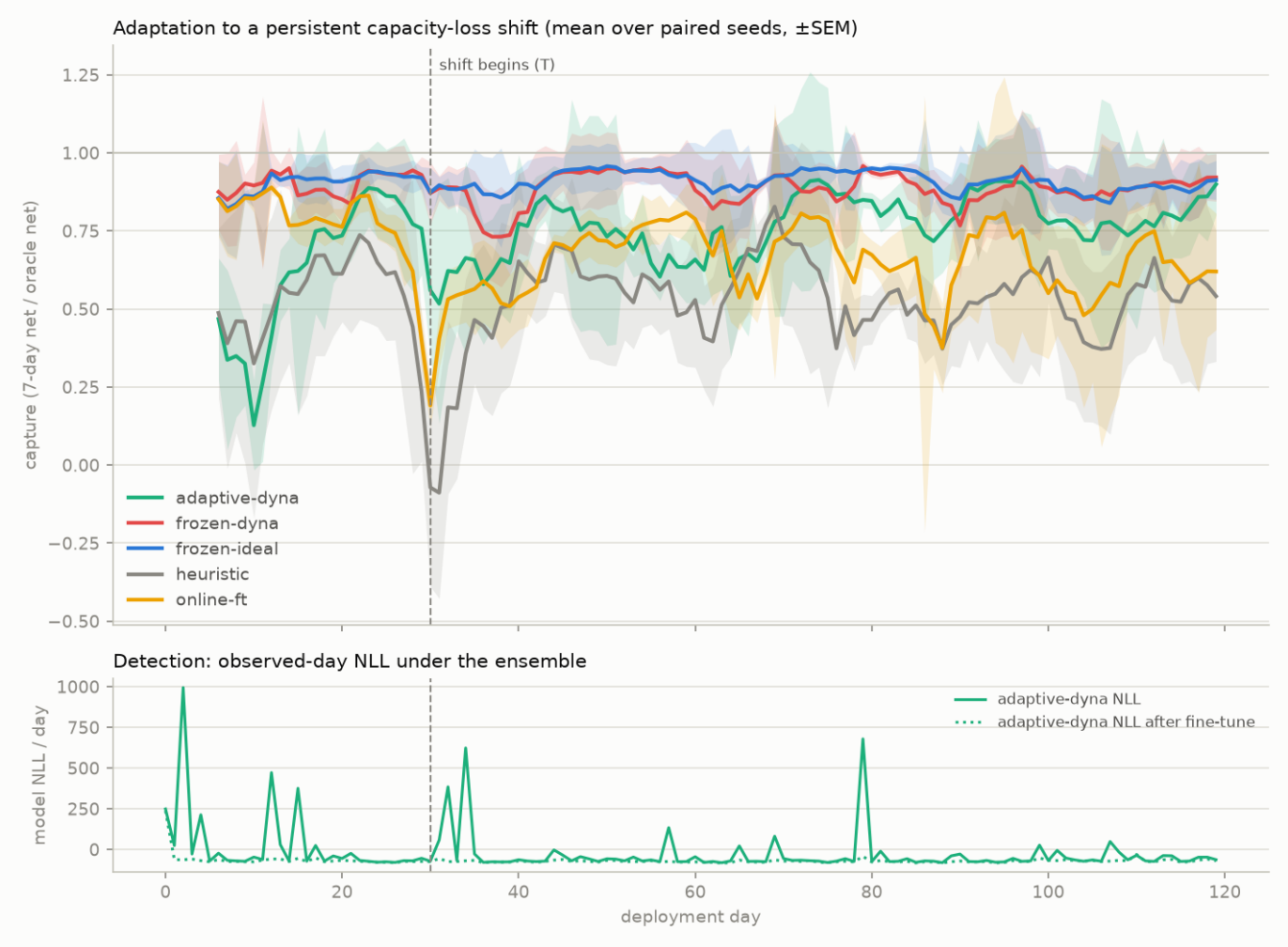

Fig. 3. Capture curves and NLL detection trace, capacity-loss shift, mean over 10 paired seeds (±SEM).

days, a policy trained only on calm data holds up when a shock it has never seen arrives, keeping 58 to 90% of what the oracle makes on the same shocked days. That is a floor rather than a solution, but it comes without any shock training.

Adaptation is where it breaks, and not for the reason we expected (Section VII-D). The difficulty isn't noticing that the market has moved. It's that refitting the model every night on a rolling window quietly degrades it, even while the market is still stationary. The model leaves its own calibration band on the third night, 27 days before the shift arrives. Keeping a model trustworthy while it goes on learning turns out to be a harder requirement than getting it right once. A stricter gate doesn't solve this because a gate can only reject a bad refit, and rejecting every refit is the same thing as freezing the model. What the loop needs is a refit that holds on to what the original year taught it while it absorbs new days, instead of drifting away from both (Section VII-D1).

The battery-side study results in a similar direction where learning the battery instead of assuming it gives no sample-efficiency gain over a plain network, and the wrong-constraints ablation shows why. The physics residual only carries information when data is plentiful, and goes inert below $N{=}50$, exactly where it was meant to help (Section VII-E1). Its zero rate of impossible predictions isn't a result but a restatement of its output clamp, which any network can use. The honest verdict is that the physics prior isn't practical on this problem. And model-free online learning is the weakest option throughout, taking years to converge from scratch and coming out worst after a shift. The through-line is that it's the world model's trustworthiness, not raw interaction, that carries this approach.

B. Limitations

Everything here is simulation, a synthetic market and battery, so none of it transfers to a real deployment without first validating against measured price and cell data. The price-taker assumption does double duty, justifying both the decomposed world model and the hindsight oracle, and would weaken for storage big enough to move prices. Every quantitative result also depends on a single setting of the world: one European heating-dominated calibration, one price cap ($1000/MWh), one battery cost model ($250/kWh replacement, 5,000-cycle

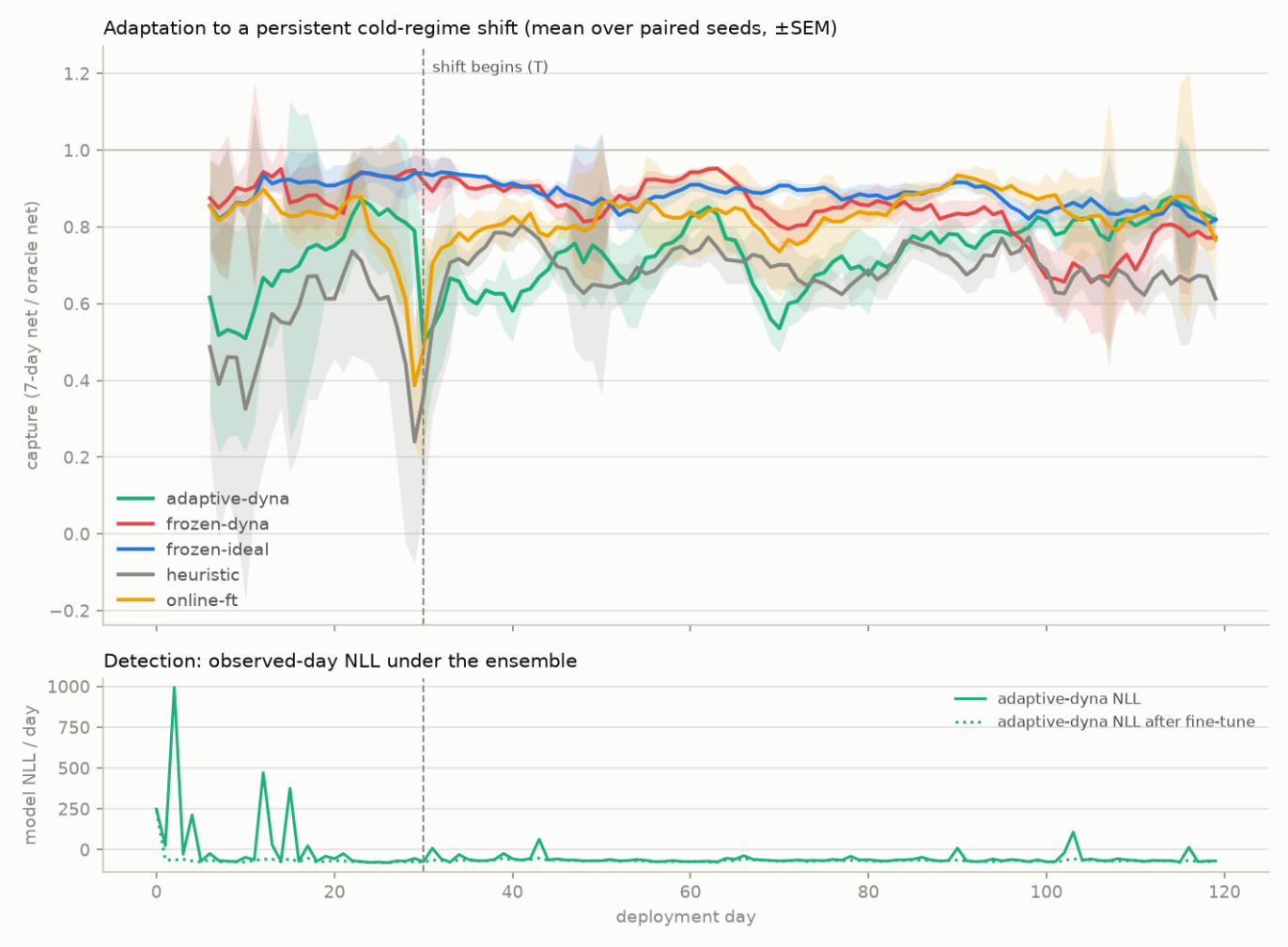

Fig. 4. Capture curves and NLL detection trace, cold-regime shift, mean over 10 paired seeds (±SEM).

life), and three hand-picked shift regimes rather than continuous or compound drift. The *sizes* of the reported gaps are therefore specific to this setting; what holds up is the orderings and the mechanisms behind them. Finally, the zero-infeasible-request guarantee of Section VII-B covers only planner-sourced data collection during training. The trained policy itself requests infeasible actions on over 90% of deployment steps, which the follow-up measurement attributes to benign SoC-bound parking, but whether a real battery-management system tolerates thousands of refused pushes a year is a deployment question the simulator can't settle. On-asset safety should be read as solved for training, characterised but not certified for deployment.

C. Future Directions

The cheapest next step is to close the missing arm in Section VII-A: train a Dyna policy inside an ensemble the gate *rejected*, such as the $D{=}90$ fit whose invented spread is more than double the real year's and whose winter and evening patterns come out with the wrong sign. That requires only an override of the guard that refuses ungated models, plus one training run, and it would turn the quality-over-quantity claim from a one-sided demonstration into a two-point measurement.

The clearest substantive next step is to repair the nightly refit (Section VII-D1). Because the ensemble drifts on stationary data, the fix belongs in the refit itself rather than in the test wrapped around it: refitting from scratch on the pooled history each night instead of fine-tuning from the previous night's weights, anchoring more heavily on pre-deployment days, or scoring the refit against the calibration statistics rather than likelihood alone, then re-running the three-shift grid. On the battery side (Section VII-E), the question worth keeping isn't about physics priors at all but whether constraining a surrogate's outputs to be physically possible protects a downstream policy from model exploitation, which needs no physics-informed variant to test. A stronger claim about physics priors would need a harder dynamics problem than a smooth three-dimensional step map; the successes reported on real cells (Wang et al., 2024) come from noisy, partially observed measurements, so the negative here bounds that claim rather than contradicting it.

TABLE VIII

WRONG-CONSTRAINTS ABLATION. HELD-OUT POOLED RELATIVE MAE, UNIFORM COVERAGE, MEAN ± STD OVER TEN FIT SEEDS, IDENTICAL TRAINING DRAWS AND TEST SET TO TABLE VII. SOFT−CORRECT IS THE TUNED SOFT-RESIDUAL PINN; SOFT−ETA−TRUE FREEZES EFFICIENCY AT ITS TRUE VALUE (A CONTROL FOR FREEZING ITSELF); THE REMAINING THREE ARMS FREEZE OR FLIP A PHYSICAL LAW TO A WRONG VALUE. **BOLD** MARKS THE BEST POINT ESTIMATE IN EACH ROW. THE CORRECT-VERSUS-WRONG GAP IS LARGE AT HIGH BUDGETS AND VANISHES BELOW N=50, WHERE THE RESIDUAL STOPS INFLUENCING THE FIT AT ALL.

N	mlp	soft-correct	soft-eta-true	soft-eta-wrong	soft-cap-wrong	soft-cal-wrong
500	0.018 ± 0.003	0.014 ± 0.002	$\mathbf{0.014 \pm 0.002}$	0.070 ± 0.004	0.071 ± 0.002	0.022 ± 0.002
200	0.034 ± 0.003	0.032 ± 0.003	$\mathbf{0.031 \pm 0.003}$	0.086 ± 0.005	0.082 ± 0.003	0.036 ± 0.003
100	$\mathbf{0.043 \pm 0.003}$	0.065 ± 0.006	0.065 ± 0.007	0.104 ± 0.005	0.100 ± 0.009	0.067 ± 0.008
50	$\mathbf{0.076 \pm 0.006}$	0.170 ± 0.020	0.166 ± 0.018	0.188 ± 0.018	0.158 ± 0.010	0.171 ± 0.019
25	$\mathbf{0.147 \pm 0.009}$	0.187 ± 0.016	0.194 ± 0.018	0.206 ± 0.012	0.217 ± 0.021	0.188 ± 0.019
15	$\mathbf{0.358 \pm 0.014}$	0.521 ± 0.011	0.523 ± 0.008	0.538 ± 0.011	0.539 ± 0.021	0.522 ± 0.012
10	$\mathbf{0.454 \pm 0.029}$	0.504 ± 0.013	0.505 ± 0.016	0.527 ± 0.018	0.494 ± 0.009	0.504 ± 0.012

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APPENDIX A
GENERATIVE AI TOOLS: STUDENT ACCOUNTABILITY STATEMENT

Student Name: Nicholas Lai Jin Yung
Project Title: A Validated Learned Market Model for Sample-Efficient Battery Storage Arbitrage
Supervisor Name: David Mguni

A. Have you used Generative AI tools for your project?

YES

B. AI tools used

AI Tool	Version	Publisher	Access URL
Claude Code (Claude Opus)	Opus 5 (`claude-opus-5`)	Anthropic	https://claude.com/claude-code
[ADD ANY OTHER TOOLS USED]			

C. How, where, and why these tools were used

Project Section	Description	Justification
Software development: simulator and environment (`energy_storage/`)	AI-assisted code generation, refactoring and debugging of the market simulator, battery model, Gymnasium environment and optimisation oracles.	Design decisions and problem formulation were mine. All behaviour is pinned by a test suite of 84 tests, including the statistical calibration contract of Table I, which I specified and verified.
Software development: experiment scripts (`scripts/`)	AI-assisted authoring of the training, measurement and plotting scripts that produce each table and figure.	Every reported number was regenerated by executing these scripts. Runs are seeded and reproducible, and the estimators were specified by me.
Results and analysis (Section VII)	AI-assisted computation of error margins (leave-one-out jackknife and standard deviations) and formatting of summary tables from stored experiment outputs.	Choice of estimator and experimental design were mine; outputs were checked against the raw per-seed data. No result was reported that the code did not produce.
Dissertation prose (Sections II, III, IV, VIII)	AI-assisted copy-editing and condensing of prose I had already drafted, to reduce length and improve clarity.	No new technical claim was introduced by the tool. All claims, results and interpretations are my own and were verified against the experimental outputs.
Related Work (Section II)	[CONFIRM AND EDIT: describe exactly how AI was used here, e.g. shortening summaries of papers you had already read and selected.]	[CONFIRM: state explicitly whether any cited work was suggested by the tool, and confirm you read every source you cite.]
Supporting material (`README.md`)	AI-assisted drafting of the reproduction guide mapping each script to the result it produces.	All commands were verified by execution before submission.

D. Generative AI Student Declaration

- I have declared all uses of AI tools (e.g., coding assistants, text generators, data analysis tools) in this project.
- I have reviewed and verified all AI-assisted content to ensure accuracy and suitability.
- I take full responsibility for the originality and integrity of my project submission.
- I have referenced all sources, including those suggested or produced with AI support.
- I have not used AI in ways that undermine academic honesty (e.g., plagiarism, unverified code submission).
- My use of AI was supplementary and did not replace my own independent learning and problem-solving.

Signed: _____ Date: _____